



Requirement Specification

Equity Lens v0.2

Post-prototype status review

Document status: Draft, internal review

Date: September 2026

Supersedes: Requirement Specification – Equity Lens v0.0 (July 2026)

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1. Purpose of this document

Equity Lens is a transparency tool that helps selection committees detect potential bias patterns before final decisions are made. It does not rank candidates, recommend outcomes, or record votes against a voter's identity. Its purpose is to make invisible bias patterns visible so that committees can apply informed human judgment with greater confidence.

This document supersedes Requirement Specification – Equity Lens v0.0 (July 2026), which recorded the requirements gathered at the Equity Lens co-creation workshop at the first UN Global Dialogue on AI Governance. That document is preserved in full below; nothing in it has been rewritten. What is new here is a status column and an implementation note against every requirement, recording what a working prototype has actually built, what it has built only partially, what it deliberately declined to build and why, and what remains untouched.

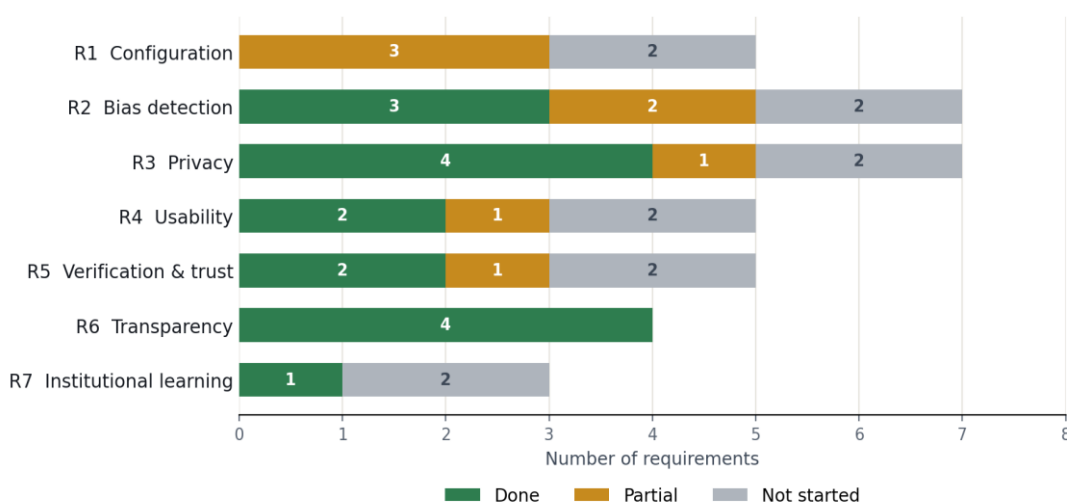
This is a living document, not a certification. A status of “Done” means the requirement is implemented and testable in the live prototype today, not that it has been formally verified, piloted, or reviewed by an external party.

How to read the status column

Status	Meaning
Done	Implemented in the live prototype and can be demonstrated today.
Partial	Some part of the requirement is implemented; a meaningful part is not. The notes column says which.
Not started	No implementation exists yet. This includes items deliberately deferred, with the reason given in the notes column.

Status at a glance

Requirement status by area (16 done · 8 partial · 12 not started, of 36)



R1 — Context-sensitive configuration

Requirements for adapting the tool to different kinds of selection.

ID	Requirement	Status	Notes
R1.1	The tool shall allow committees to configure which bias flags are active for their specific selection context.	Partial	Two-tier model implemented: four checks (gender balance, institutional concentration, career interruptions, impact beyond citations) always run and have no switch. Two further checks (familiarity, career-stage productivity) can be toggled per session. Not yet an open catalogue of configurable flags.
R1.2	The tool shall support different selection types (professorships, junior hires, speaker invitations, awards) with context-appropriate criteria.	Partial	Four selection-type profiles are live: speaker invitation, award, junior hire, senior hire/professorship. Each sets its own form fields and default flag configuration. Professorship currently shares the senior-hire profile rather than having its own.
R1.3	Committees shall be able to adjust weights for different metrics.	Not started	Deliberately deferred.
R1.4	The tool shall allow multiple selection rounds (e.g., shortlisting).	Not started	No round or stage concept exists. Each session is a single flat vote.
R1.5	The tool shall allow committees to include soft-skills criteria in candidate profiles, if relevant to the selection context.	Partial	For junior and senior hire profiles, teaching portfolio, supervision record, third-party funding, and international mobility are collected and shown in the candidate table.

R2 — Bias detection (what to flag)

Requirements for the checks the tool runs against a candidate pool.

ID	Requirement	Status	Notes
R2.2	Flag institutional concentration (when $\geq 40\%$ of nominees from one institution).	Done	Implemented with a refinement beyond the literal spec: below three candidates, a percentage carries no information (any split is 50/50 or 100/0), so the flag only fires below that size when every candidate shares one institution. At three or more candidates the 40% rule applies as specified, and ties at the top name every tied institution rather than one at random.
R2.3	Flag familiarity bias (candidates with ≥ 2 prior invitations in 3 years).	Done	Threshold matches the specification exactly.
R2.4	Flag career-stage mismatch (comparing junior to senior unfairly).	Partial	The career-stage productivity flag (m-quotient vs raw h-index) addresses part of this. There is no explicit flag for a junior candidate being directly compared against a senior one within the same pool.
R2.5	Flag network effects (candidates from same institute, advisor or research group).	Not started	No advisor or research-group field is collected, so this cannot be checked yet.
R2.6	Flag citation bias — note that h-index itself may carry gender bias.	Partial	Implemented as an explanatory note rather than a metric adjustment. A literature review during development found that in physics, the h-index gap between women and men appears to run through productivity and career length rather than through citation rate itself — women in the pooled evidence are cited at parity or above once output and career length are controlled for. On that basis, a per-field multiplier on h-index was considered and deliberately rejected: it

ID	Requirement	Status	Notes
			would risk asserting a precision the evidence does not support, and would apply a different standard by gender, which raises its own fairness and legal questions. The tool instead carries a note beside the h-index field stating the documented gap and its likely source, and leaves the number itself untouched.
R2.7	Field-adjustable metrics — allow different h-index interpretations for different fields.	Not started	Not built, and now understood to interact directly with the R2.6 decision above: any field adjustment to a citation metric needs the same scrutiny that led to rejecting a gender-based one.
R2.8	The tool shall offer double-blind mode as an optional setting (masking name and gender) in preventive mode, not as a default or mandate.	Done	Implemented as a checkbox the chair sets before publishing. When enabled, member-facing candidates are renamed "Candidate 1", "Candidate 2" etc. in a shuffled order, gender is withheld, and key publications are withheld (a title and journal can identify an author to a specialist committee). In a normal session key publications are visible to the whole committee, so members weigh the actual work rather than the metrics alone. Institution, career stage and quantitative metrics remain visible in both modes. The chair is never blind. The interface states plainly that this reduces identifiability, it does not guarantee anonymity in a small field.

R3 — Privacy & data protection (DSGVO/GDPR compliance)

Requirements aligned to specific GDPR articles.

ID	Requirement	Status	Notes
R3.1	Never store individual votes — only aggregated totals. (Art. 5(1)(c), data minimisation)	Done	Implemented structurally rather than procedurally. Seat tokens (hashed) and per-candidate vote counts live in two separate database tables with no column joining them. The pairing between a member and a ballot is not merely undisplayed — it was never written, so it cannot be recovered by the chair, or by anyone with database access.
R3.2	Candidate names shall be hashed after 30 days. (Art. 32)	Not started	No automatic expiry or hashing job exists. The candidate list is encrypted client-side before it reaches the server, but the ciphertext is retained indefinitely on the server until the chair manually presses "Delete all session data." Auto-deletion of the encrypted roster at the moment voting completes, plus expiry of abandoned sessions after a fixed window, is the recommended next step for this requirement.
R3.3	Users shall be able to delete all their data. (Art. 17, right to erasure)	Done	A "Delete all session data" control removes the session's ballots, seat records and encrypted roster from the server, and clears the chair's local draft.
R3.4	Data shall be encrypted at rest (AES-256) and in transit (TLS 1.3). (Art. 32)	Done	The candidate roster is encrypted client-side with AES-256-GCM before it leaves the chair's browser; the decryption key travels to members only inside the URL fragment, which browsers never transmit to a server. All network traffic runs over Cloudflare's TLS.
R3.5	Differential privacy ($\epsilon=0.1$) for all aggregated outputs, where applicable. (Art. 25)	Not started	Judged, during development, to be mis-scoped as originally written: adding calibrated noise to a five-member committee's vote tally would change the declared winner, which defeats the purpose of a vote.

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			The recommended rescoping is to apply differential privacy to the cross-committee benchmark aggregates (see R7.3) once that feature exists, where a five-record minimum cell size is already enforced as a first-line protection.
R3.6	Minimum cell size (fewer than 3 individuals = data suppressed).	Done	The in-session aggregate statistics panel (gender counts, institution count) is withheld and replaced with an explanatory note whenever a pool has fewer than three candidates. Separately, the server-side archive (see R7) enforces its own minimum cell size of five records before publishing any cross-committee breakdown.
R3.7	Users shall be able to opt out of data being used for AI training. (Art. 7, Recital 42-43)	Partial	A checkbox — "Do not offer this session for research or AI training" — is presented when a session opens, and the choice is recorded for that session. It does not yet gate anything, because the archive-contribution flow it is meant to control is itself not yet live in the interface (see the architecture gaps noted under R7). The control exists ahead of the feature it governs, by design, so the opt-out will already be honoured once contribution goes live.

R4 — Usability & workflow

Requirements for how the tool feels to use.

ID	Requirement	Status	Notes
R4.1	5-minute rule — total committee time from login to vote ≤ 5 minutes.	Not started	Not measured. The workflow (open session, enter a small pool, publish, vote) is short in principle but has not been timed with real users.
R4.2	Automated candidate upload — support CSV or similar bulk upload.	Not started	Candidates are entered one at a time through the form. No file import exists.
R4.3	No training required — intuitive interface with clear labels.	Partial	Labels and helper text have had several rounds of revision for clarity, but this has not been tested with committee members unfamiliar with the tool.
R4.4	Plain-language flags — each flag explains what it means and suggests an action.	Done	Every flag carries a one-sentence finding and a one-sentence suggestion, written for a non-technical reader.
R4.5	Multi-device support — works on desktop, tablet, and mobile.	Done	Implemented and verified. The interface collapses to a single column on phones, buttons meet a 44px minimum touch target, inputs use 16px text so mobile browsers do not zoom on focus, and the candidate detail table is replaced below 720px by one stacked card per candidate rather than a table that must be scrolled sideways. A fixed navigation bar appears once a session has several sections, so a committee can jump between candidates, bias check and ballot without scrolling past everything. Verified at 320px, 390px and 1280px with zero horizontal overflow.

R5 — Verification & trust

Requirements for the tool being trustworthy to inspect, not merely to use.

ID	Requirement	Status	Notes
R5.1	The tool itself shall be auditable — code open-source and inspectable.	Done	Resolved. The repository is public at github.com/AKC-Equity-Lens/equity-lens under the MIT Licence, with the copyright held by Ruzin Aganoglu. The code has no build step and no runtime dependencies, so what a reviewer reads is exactly what runs.
R5.2	The tool shall support third-party verification of its bias detection logic.	Partial	Now possible, but not yet done. Public release under an open licence removes the access barrier, and the bias engine is plain commented JavaScript with no dependencies, which makes independent review straightforward. What is still missing is an actual third-party review having taken place. CONTRIBUTING.md asks explicitly for challenges to individual indicators and thresholds.
R5.3	The tool shall provide transparency on its own limitations — what it does not detect.	Done	A "Limitations" panel is shown in the interface itself, not in external documentation, stating what the tool cannot detect, what it cannot verify, and the specific limits of double-blind mode.
R5.4	Bias flags shall include confidence indicators (e.g., "strong signal" vs. "weak signal").	Not started	Deliberately deferred. Labelling a flag's confidence is itself a statistical claim; building the labelling mechanism is straightforward, but deciding what evidence earns a "strong" label is not, and doing it without a defensible rule risks lending false authority to a pattern found in three or four candidates.
R5.5	The tool shall allow committees to dismiss flags with justification (optional).	Not started	Not built. Recommended as a higher-priority item than R5.4 once building resumes, since dismissal is what should replace the ability to simply switch a check off, and gives the tool a record of when and why a committee disagreed with a flag.

R6 — Transparency & accountability

Requirements for what the tool shows and what it refuses to do.

ID	Requirement	Status	Notes
R6.1	The tool shall provide a dashboard showing aggregated, anonymised patterns.	Done	The bias-check dashboard shows pool statistics and flags computed from the candidate pool, never from an individual candidate in isolation.
R6.2	The tool shall never influence committee choices — it only informs.	Done	Enforced by omission: no ranking, scoring, or recommendation exists anywhere in the interface, and this was treated as a hard constraint throughout development rather than a stated intention.
R6.3	The tool shall support export of anonymised session data for institutional learning.	Done	A chair can download a local JSON summary of a completed session (candidate count, gender balance, institution count, flag types and severities). This is a local file export, distinct from the cross-committee archive described under R7.
R6.4	The tool shall include a privacy notice explaining what data is collected and why.	Done	A privacy notice is built into the interface itself, stating what is held on the chair's device, what is held on the server and for how long, where it is physically stored, and who is responsible for the candidates' data.

R7 — Committee self-awareness & institutional learning

Requirements for learning across sessions and across committees.

ID	Requirement	Status	Notes
R7.1	The tool shall provide committees with aggregated, anonymised feedback on their own decision patterns over time.	Not started	The server-side storage for this (an "archive" with a strict, whitelisted, anonymous record schema, minimum cell size, and no reference back to any voting session) has been built and deployed. The client-side contribution flow that would let a chair actually send a record to it has not: its interface exists in the page but is not yet connected or shown. See the architecture gaps section below.
R7.2	The tool shall not identify or track individual committee members' votes — only aggregate patterns.	Done	Same structural guarantee as R3.1.
R7.3	The tool shall provide comparison to anonymised benchmarks (e.g., "compared to similar committees...").	Not started	Depends entirely on R7.1 being wired up and on enough sessions contributing records to clear the five-record minimum cell size. This is the feature intended to make contribution worthwhile to a chair, not merely a donation: "your committee selected women 20% of the time from a pool that was 40% women; across comparable committees the figure is 35%."

Beyond the original specification

Building the prototype surfaced problems the workshop had no way to anticipate. These items are not tied to a requirement ID because none existed for them; they are recorded here so the next round of review can decide whether they deserve one.

- Candidate-list locking: once a chair publishes a slate, it locks — no candidate can be added that the committee never saw on its ballot. A controlled "reopen to edit" is available only before the first vote is cast, and requires republishing so members are never left voting on a stale list.
- Session recovery: a chair's browser refresh no longer loses the candidate list, seat links, or configuration — session state is held per browser tab and restored automatically.
- Institutional-concentration false positive fixed: the original 40% rule, applied literally to a two-candidate pool, could flag a perfectly diverse pool as concentrated. A minimum-pool rule now prevents this (see R2.2).
- Gender recorded as a three-way field (woman / man / diverse) throughout the dashboard, the export, and the interface, rather than a binary.
- h-index lookup fixed: the original Scopus deep-link had been retired by Elsevier and failed silently. It now opens Scopus's current author-search form and copies the candidate's name to the clipboard, with an on-screen note that Scopus, Web of Science and Google Scholar give different values for the same person.
- Key publications are now shown to the whole committee in a normal session, not only to the chair. Previously the field was collected and transmitted but rendered for nobody but the chair, which meant members voted on metrics without seeing the work behind them. In double-blind mode the field remains chair-only.
- Automatic version handling: every page carries a visible build stamp, loads its script pinned to that build so a new page can never run cached old logic, and checks for a newer build on each load, refreshing itself once if it finds one. This replaced a fragile manual workaround (appending ?v=2, ?v=3 to the address) which did not work at all for committee members, since seat links are generated by the tool and carry no version parameter — members were silently being served stale pages.
- QR codes for seat links, generated by an encoder written into the project rather than loaded from a content delivery network — a seat link is a credential, and the privacy notice states that no third-party scripts are loaded. A presenter mode shows one large code at a time, so seats can be handed out by scanning in a room rather than by message. The encoder was verified against a reference implementation and its output decoded with an independent scanner across QR versions 1 to 11.
- Readability pass across the whole interface: nothing below 13px, body text at 16px, form labels at 15px, and darker grey text where the previous shade fell below normal contrast thresholds.
- Repository governance published alongside the code: MIT licence, contributing guide stating the design constraints a change must respect, code of conduct, security policy with a severity ordering specific to ballot secrecy, and CITATION.cff so the software can be cited.

Known architecture gaps

These are specific, concrete items blocking R7 (institutional learning) and parts of R3 (privacy) from moving past their current status. They are listed separately from the requirement tables because they are implementation facts, not requirement text.

- The "Contribute to the open dataset" panel exists in the page's markup but is never shown or wired to a button handler. It should be treated as not yet built for planning purposes, even though its server-side counterpart (the archive) is live.

- The archive's server-side flag whitelist does not yet include "impact-beyond-citations", which the bias engine added after the archive schema was written. The two need to be reconciled before any real record could be accepted.
- The archive's selection_type field currently accepts award / speaker / hire / other. The live tool now distinguishes junior hire from senior hire. The schema needs a matching update, or an explicit decision to collapse both into "hire" for archive purposes.
- No automatic deletion or expiry job exists yet on the server (see R3.2). Data is retained until a chair manually erases it.
- The hosting and legal model is not yet settled for large volume of users.